The first edition of the conference was held in 2025. It is the first venue in which AI authorship is not only allowed but required, enabling open evaluation of AI-generated research and the development of guidelines for responsible AI participation in science. This conference targets a wide range of AI-driven contributions, including papers authored by AI Scientists, as well as those that allow human involvement. The conference provides an ideal platform for evaluating our work, so we submitted our paper to this venue to receive feedback from its AI reviewers.

5.2 AI Reviewer in Agents4Science.

The AI reviewers used in Agents4Science are based on GPT-5 (OpenAI, 2025a), Gemini 2.5 (DeepMind, 2025), and Claude Sonnet 4 (Anthropic, 2025). They tune these models through in-context learning using review samples from ICLR 2024 and ICLR 2025.

5.3 Review Results.

We submitted papers generated by the earlier version of Jr. AI Scientist. Although these are not identical to the newer papers in this study, the reviews discussed here mainly apply to the newer papers as well. We summarize below the representative comments from the submitted reviews.[1] In terms of strengths, the reviewers generally noted that the work is technically sound, includes comprehensive ablation studies, and is clearly presented. As for the weaknesses, we identified four key issues that we consider particularly important, as summarized below.

Weakness1 :

Limited Improvement over Baselines.

This comment is valid. While Jr. AI Scientist achieved higher scores than the baseline, the performance gap is not significant enough to claim a substantial improvement. To address this limitation, it would be necessary to increase the number of experimental trials and explore more innovative search strategies for selecting experimental nodes.

Weakness2 :

Moderate Novelty and Incremental Contribution.

This observation is also reasonable. Since Jr. AI Scientist is designed to build upon a given baseline, a certain degree of incremental progress is inevitable. Achieving more innovative ideas would likely require human intervention during the idea generation phase.

Weakness3 :

Insufficient Experiments. No Comparison with Other Methods.

We agree that comparisons limited to the baseline are not sufficient. However, expanding the comparative methods would require appropriate selection of comparison methods and accurate reproduction of them, which remain beyond the current level of autonomous AI Scientists. Therefore, human intervention would also be necessary in this part.

Weakness4 :

Shallow Theoretical Justification.

[1]Detailed reviews are available at the following URLs:
LoCoOp: `https://openreview.net/forum?id=x7qlIDcw0P`
GL-MCM: `https://openreview.net/forum?id=AzOkqwsTXo`
Min-K%++: `https://openreview.net/forum?id=L5gDfr4GdF`

This comment is fair. Jr. AI Scientist follows an experimental, performance-driven design that repeatedly edits and improves code until it surpasses the baseline. Therefore, it does not include a mechanism to theoretically validate why a particular modification works. As a result, some successful solutions may have been discovered only by chance, and their effectiveness might not generalize to other datasets.

For these reasons, our submission was rejected from the Agents4Science conference. However, we would like to emphasize that most of the accepted papers at this venue involved human intervention, so the rejection does not necessarily indicate that the capability of our AI Scientist is low. The feedback we received clearly highlights the current limitations, and we believe these points will serve as important directions for the future development of AI Scientists.

6 Authors Evaluation

We conducted an internal review of the generated papers with authors. Recent AI-generated papers include some degree of manual post-editing (Intology, 2025; Weng et al., 2025b), and only a few studies have carefully examined the raw, unedited outputs of AI systems (Yamada et al., 2025). However, evaluating the raw, unedited outputs is essential for accurately understanding the current limitations of AI Scientist systems.

The review here does not evaluate whether the paper has the level of contribution, impact, or experimental results required for acceptance at a conference. Instead, our review focuses on whether the writing contains misinterpretations of results, incorrect methodological descriptions, inaccurate citations, or hallucinations. Therefore, we cross-check the manuscripts against the actual code and experimental results to precisely identify such issues. The issues of our review are summarized as follows. More detailed reviews for each paper are provided in the supplementary material.

Positive aspects are that none of these papers contained citations to non-existent works or developed invalid methods, such as test-data-leaking methods. The issues in these papers are as follows:

Issue1 :

Frequent Irrelevant Citations.

We found that there are some irrelevant citations in these papers. This issue arises when adding new BibTeX entries that are not included in the baseline papers. (The reason for this is discussed in detail in Writing Risk 2 of Section 7.)

Issue2 :

Ambiguous Method Descriptions.

We found that while the method descriptions are generally accurate, they still contain ambiguities. For example, in the LoCoOp extended paper, the parameter appearing around Line 156 is not clearly explained, making it difficult to fully understand the method. Similarly, in Min-K%++ extended paper (Lines 126–129), although the corresponding code exists, the process is implemented as an optional component and is not actually utilized. This occurs because the coding agent makes numerous modifications during Stage 2 in the Experimental Phase, increasing code complexity. This suggests that accurately transferring experimental code into a faithful methodological description remains an open challenge.

Issue3 :

Misinterpretation of Figure Results.

These papers include the overinterpretation of the figure results, making unsupported claims that appear plausible. For instance, in Min-K%++ extended paper (L177) and LoCoOp extended paper (L160–162), they report findings not evident from the figures. This highlights that precise result interpretation remains difficult for current AI Scientist systems.

Issue4 :

Descriptions of Auxiliary Experiments That Were Never Conducted.

We found several cases where these papers describe auxiliary experiments that were never actually conducted, such as in LoCoOp extended paper (Lines 183–184) and GL-MCM extended paper (Lines 208–213). This issue occurred even though the writing agent was explicitly instructed not to include nonexistent experimental results. This problem is especially tricky because hallucinations do not appear in the main results, which are easy to notice, but they often appear in parts like ablation or analysis. Therefore, even human reviewers might not notice them unless they carefully check the draft. Such cases illustrate that the risk of hallucination remains inherent in the current system.

7 Observed Risks During the Project

In this section, we describe the risks identified during this project. In the previous section (Section 6), we mainly identified issues related to the papers released in this study. In this section, we present various risks encountered during the development process. Sharing such risks is essential to prevent overreliance on these systems and to promote a deeper understanding of AI Scientists within the research community.

7.1 Idea Generation

Idea Risk1:

Identifying a successful idea is highly computationally expensive.

The ideas generated by AI do not always work, which holds true for human scientists. In our case, we aimed to generate one successful idea for each baseline paper. To this end, we generated approximately ten ideas and evaluated them. Some were filtered out through human review, while others did not outperform the baseline. Finally, only one idea proved to be successful.

From this perspective, more extensive validation was conducted in some recent works (Liu et al., 2025; Weng et al., 2025b). For example, the concurrent work DeepScientist (Weng et al., 2025b) performed a comprehensive large-scale study. They report that, out of approximately 5,000 unique scientific ideas generated, only 21 ultimately led to genuine scientific innovations. Our experiments require less time because our limitation analysis is effective, and our goal is modest, aiming to find one successful idea rather than exploring more successful ideas.

Validating large-scale ideas is highly computationally expensive and often infeasible for many academic laboratories. Future research will therefore focus on developing more efficient idea-pruning mechanisms, an efficient tree search algorithm for experiments, or incorporating human feedback.

7.2 Experiment

Experiment Risk1 :

Lacking domain expertise, the coding agent sometimes produces code that leads to incorrect implementations and false performance gains.

Because the coding agent is unaware of domain-specific conventions, it often improves performance in undesirable or invalid ways. This issue frequently appeared in the experiments on GL-MCM (Miyai et al., 2025b), which we describe in detail below.

Background. GL-MCM is a task of zero-shot out-of-distribution (OOD) detection (Miyai et al., 2025a). OOD detection aims to distinguish between samples belonging to a predefined in-distribution (ID) class set (e.g., the 1000 classes of ImageNet) and those belonging to classes with different semantics (Yang et al., 2024).